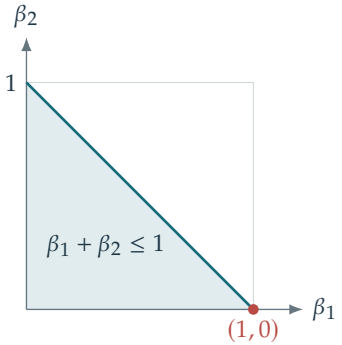
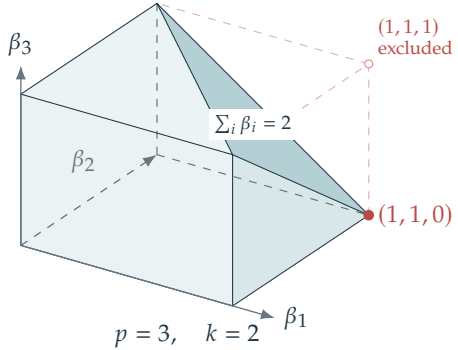
Panel 1



$p = 2, \quad k = 1$

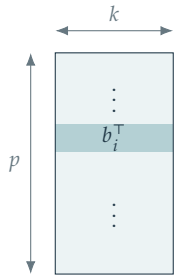


$p = 3, \quad k = 2$

Feasible region: $0 \le \beta_i \le 1$, $\sum_i \beta_i \le k$. Red dots: a maximizing vertex.

$$\lambda_1 \ge \cdots \ge \lambda_p \ge 0, \qquad \max \textstyle\sum_i \lambda_i \beta_i = \sum_{i=1}^{k} \lambda_i$$

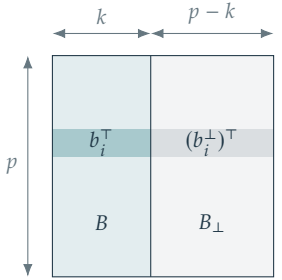
Panel 2



$B = V^\top S, \quad B^\top B = I_k$

$\beta_i = \|b_i\|^2 \ge 0, \quad \sum_i \beta_i = k$

Panel 3



$\widetilde{B} = [B \; B_\perp], \quad \widetilde{B}\widetilde{B}^\top = I_p$

$\|b_i\|^2 + \|b_i^\perp\|^2 = 1 \quad \Rightarrow \quad \beta_i \le 1$